

Supplementary Methods and Artifact Record

SpecGuard-Chem 0.1.0

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Corrected full-matrix draft, 25 July 2026

1 Release status

This supplement accompanies the results manuscript *Before the Lab Acts: Benchmarking Language Models for Constrained Compound Selection*. The corrected data, deterministic baselines, and full 546-response provider matrix are complete and audited. The fixed one-card pilot remains a staged-execution record; model comparisons use the six complete 91-card raw traces and their six zero-call post-hoc-repaired views. No historical paper-50 response or score is reported here. Those artifacts are invalid because positional split references were previously misread as dataframe labels; they remain only as immutable provenance. Tagging also remains blocked on the final paper, licensing, clean-checkout verification, and bundle checksums. Chemical-diversity extensions are deferred to a separately versioned future analysis.

2 Artifact visibility contract

Field	Visibility	Role
Task, target, endpoint, source, activity semantics	System	Assay context and higher-is-better pChEMBL identity
Support IDs, structures, descriptors, pChEMBL activity	System	Project-local evidence observed before selection
Candidate IDs, structures, permitted descriptors	System	Finite action space
Constraints and budget	System	Executable action contract
Candidate activity	Evaluator only	Retrospective utility, ranking, and regret
Raw source paths and build-only metadata	Neither	Construction provenance, not model evidence

The public JSONL and scorer JSONL share benchmark, data, source, and transform provenance. Every scorer row carries the SHA256 digest of the canonical public card. The evaluator verifies identical task sets, candidate identity and order, provenance, and binding hashes before hydration. Non-oracle systems receive a strict allowlisted projection even when legacy monolithic fixtures are used.

3 Final data identity

Artifact	SHA256
Normalized corrected records	cec651b6e97f044bf82820c465b441763cafa18a34b12361a33e79ab30faf438
Build configuration	425a40d3b64ac8398c49dfc04508e1c66522754c4e07e75e69810bc2517d9a6a
System-input cards	c18e66c726bb26f8afc3ba8422b21ec327444560d92750421f0dc44a2f393d9e
Scorer outcomes	96b5d6060e3c75dda34d835fd166fd074ca5621c18924aa0ea2714acba173ff4
Card-build metadata	d986ba96589032c59dac2dcc24cadf9aa3325616fa2a395cec682ece7220af54
Card-build audit	abb80ad110ef71d69a2cea2f09cc456399a81d431bc9e365320ab8b27064812c
Exact 546-request export	50e518893b19d4a7efd64c62e08ab94d610815f8fb7518c9af4b64ff40b6f6c5
Pre-run cost estimate	d11ce35da68e5082153be4bc57c027915ccb76cf8fb19b02e5adf767b2ab525d

An independent final rebuild produced byte-identical system input, scorer outcomes, build metadata, and build audit. A second baseline run produced byte-identical per-system traces and score directories. A second request export produced the same 546 rows and request-file digest.

4 Task inclusion and exclusions

All 100 official L0_A11 task keys were considered. The inclusion rule keeps every task with at least ten candidates satisfying the declared candidate constraints. Ninety-one tasks are included. The following tasks were excluded without outcome-based selection:

CARA task key	Feasible candidates
CHEMBL1827734_IC50	0
CHEMBL3705332_IC50	0
CHEMBL3707531_IC50	0
CHEMBL3887078_IC50	0
CHEMBL3887456_EC50	0
CHEMBL3888518_Ki	0
CHEMBL3888871_Ki	0
CHEMBL3889212_IC50	6
CHEMBL3889266_IC50	8

The included endpoint distribution is 68 IC50, 17 Ki, 5 EC50, and one Potency task. Each included task has 50 support measurements and a distinct CARA target identifier.

5 Constraints

Output validity requires exactly ten unique candidate IDs, membership in the supplied candidate pool, and support-set exclusion. Candidate eligibility requires a parseable structure, molecular weight no greater than 500, and cLogP no greater than 4.5. The configured forbidden-SMARTS list is empty. These rules are eligibility controls, not a claim of synthesis feasibility, selectivity, ADMET suitability, toxicity, or medicinal-chemistry approval.

6 Systems and frozen provider matrix

Deterministic systems comprise a random-valid selector, descriptor-desirability rules, similarity to the best observed support compound, per-task random-forest, gradient-boosting and linear support-vector regressors, and a non-deployable hidden-outcome oracle.

The provider matrix contains three conditions and two representations across 91 cards (546 requests):

- OpenAI `gpt-5.5-2026-04-23`, low reasoning, direct JSON;
- Anthropic `claude-opus-4-8`, direct JSON without extended thinking; and
- DeepSeek `deepseek-v4-pro`, thinking disabled, as checked on 16 July 2026, with the provider-retained model identifier retained in traces.

The basic interface supplies IDs, structures, molecular weight, and cLogP. The descriptor-enriched interface additionally supplies TPSA, hydrogen-bond donor and acceptor counts, and rotatable-bond count. Both receive the same action contract and an explicit instruction that pChEMBL is higher-is-better. The conservative uncached pre-run estimate is \$106.06, with maximum estimated input 158,274 tokens and a planned hard cost ceiling of \$120.

The staged pilot completed all six calls with one recorded attempt each, full provenance, and \$0.449700535 aggregate usage-derived cost; cache-only replay reproduced every score artifact. The later authorized matrix completed all 546 responses. Each of the six raw conditions has one successful provider attempt per card and complete model, response, usage, latency, request-hash, and retained content evidence. The six repaired traces cover the same 91 cards and add zero provider calls. Usage-derived token-pricing cost is 58.957 dollars for the full matrix, with complete cost coverage.

7 Deterministic post-hoc repair

Repair is computed from the same raw response and adds zero provider calls:

1. Parse and validate the raw output against the action contract.
2. Retain valid raw candidate IDs in their original order.
3. Remove duplicates, out-of-pool IDs, support identities, and infeasible candidates.
4. Fill unoccupied slots using the deterministic rules-only ranking.
5. Revalidate and store both outputs and issue sets, the source-trace digest, the repair policy, and a zero-provider-call attribution field.

Raw metrics characterize model behavior. Repaired metrics characterize the model-plus-harness system and must never be relabeled as unaided model ability.